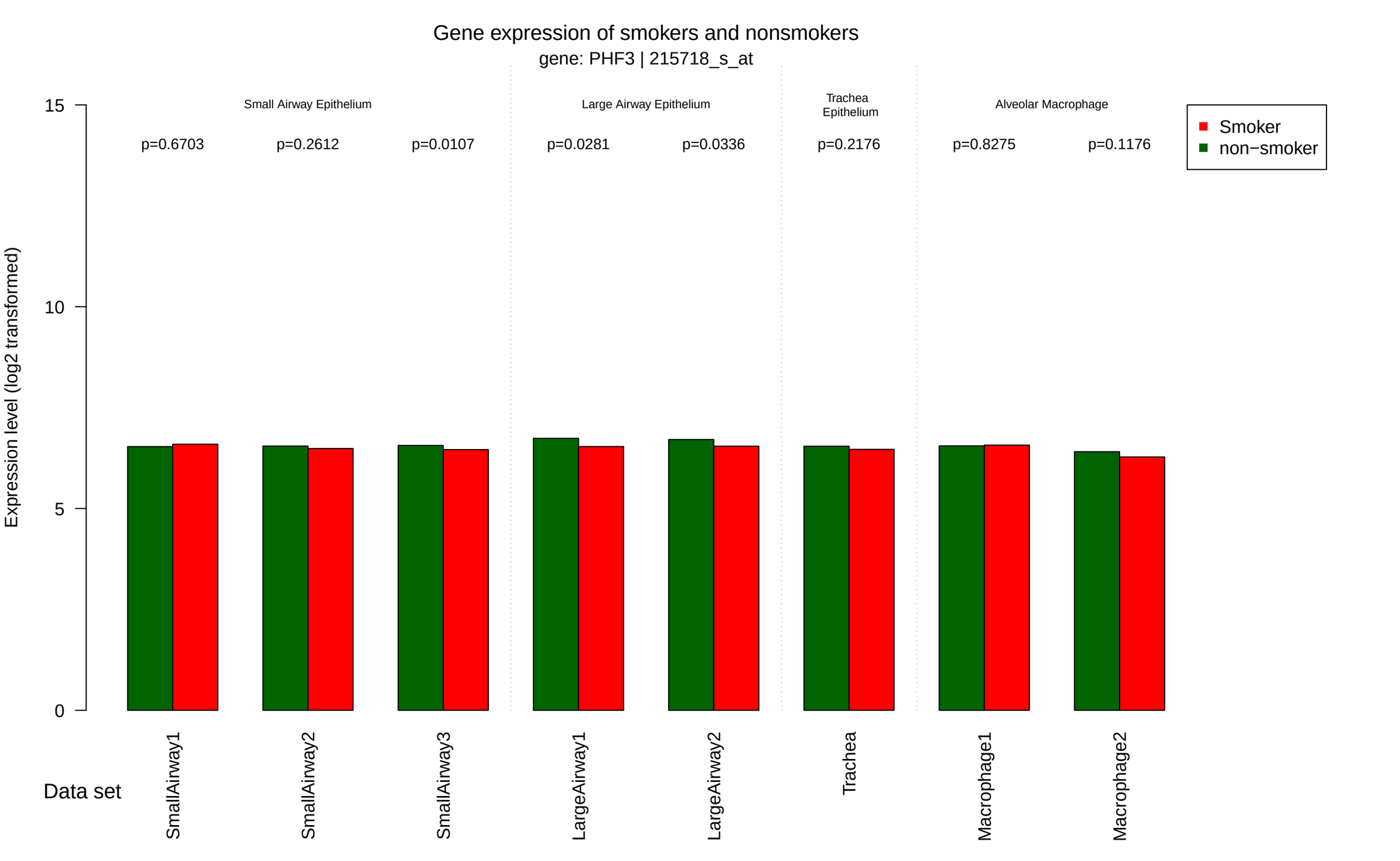
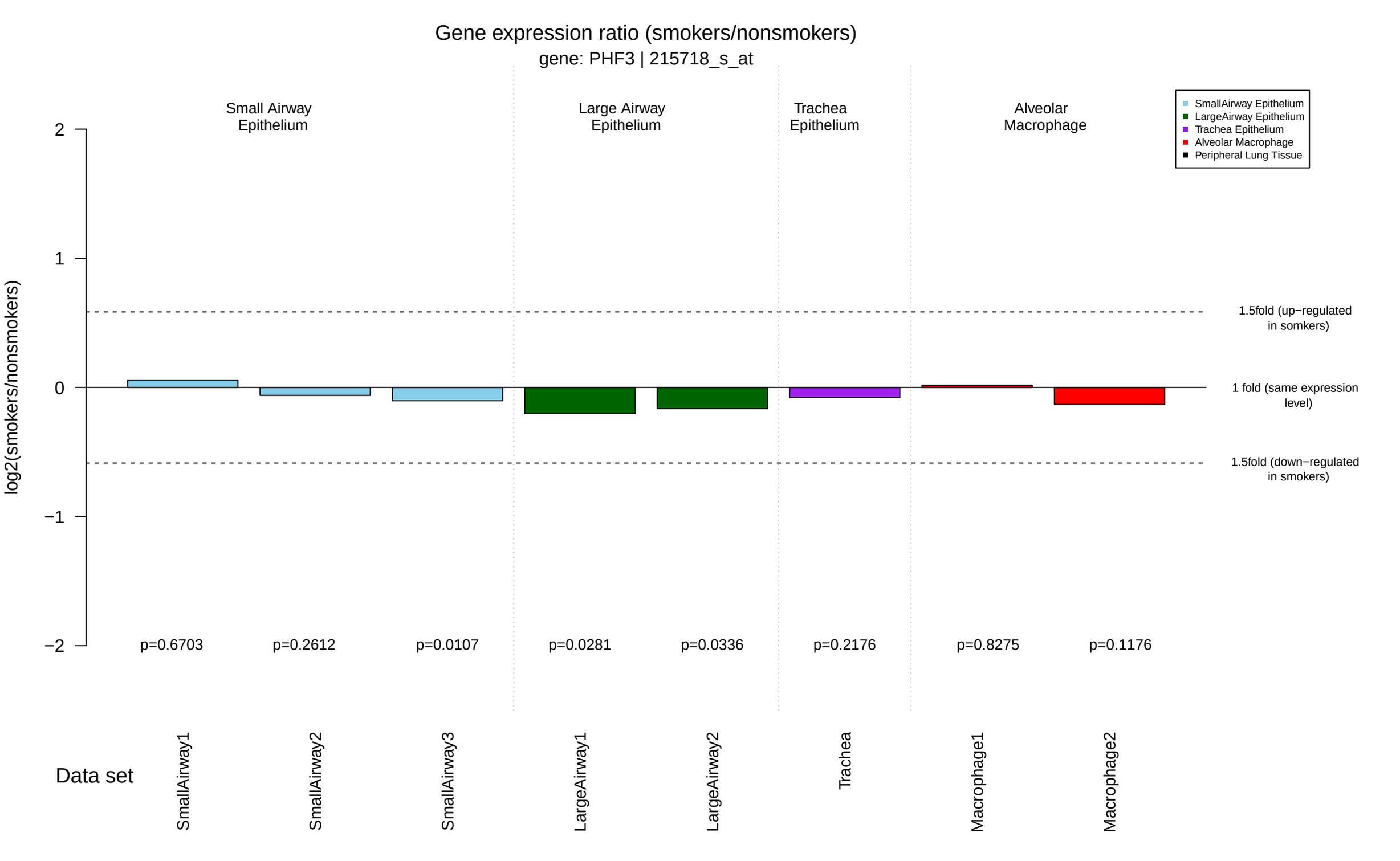
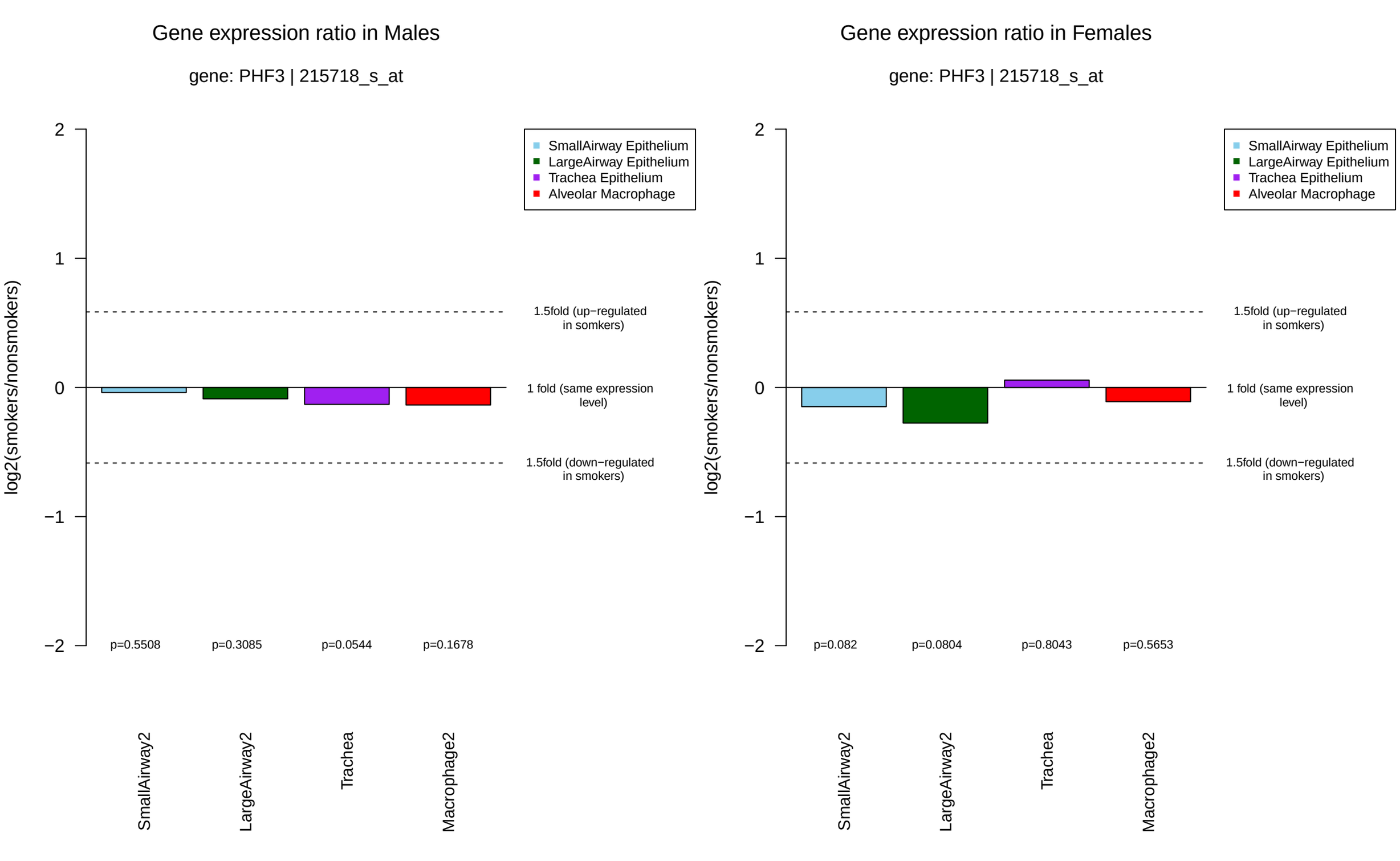
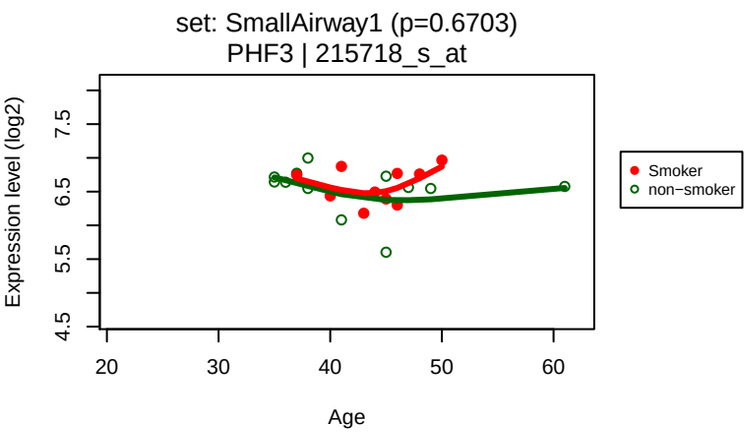
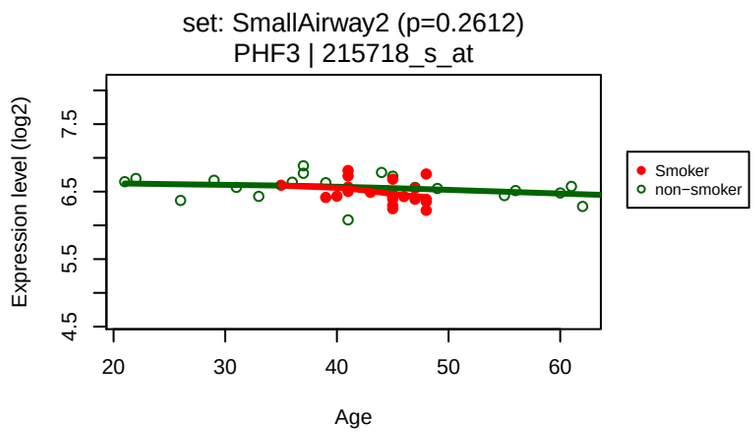
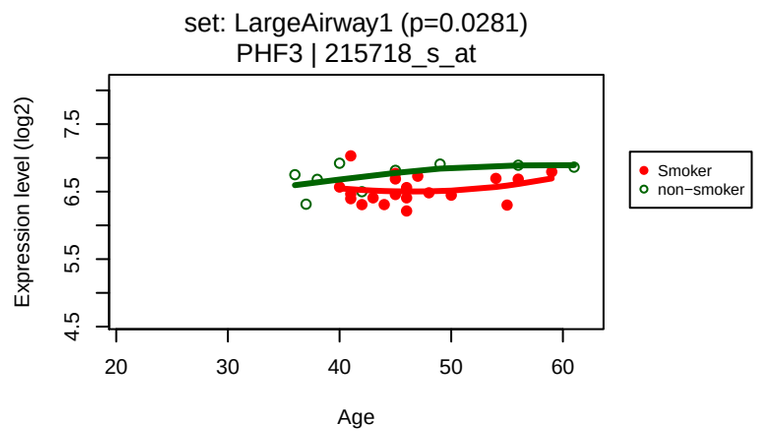

Gene expression of smokers and nonsmokers
gene: PHF3 | 215718_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6703
p=0.2612
p=0.0107
p=0.0281
p=0.0336
p=0.2176
p=0.8275
p=0.1176
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

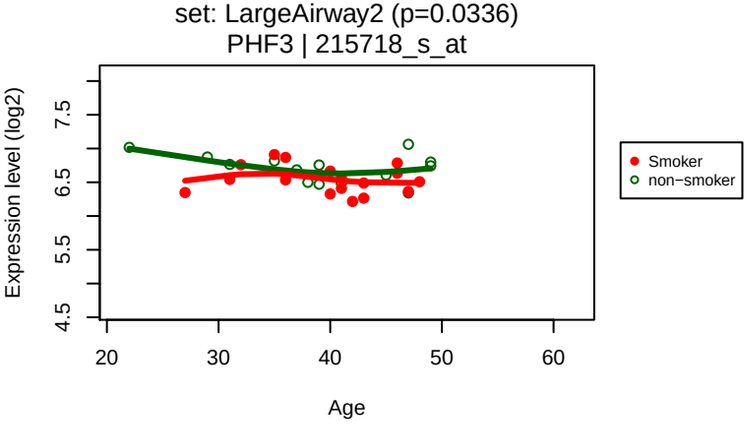

Gene expression ratio (smokers/nonsmokers)
gene: PHF3 | 215718_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6703
p=0.2612
p=0.0107
p=0.0281
p=0.0336
p=0.2176
p=0.8275
p=0.1176
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

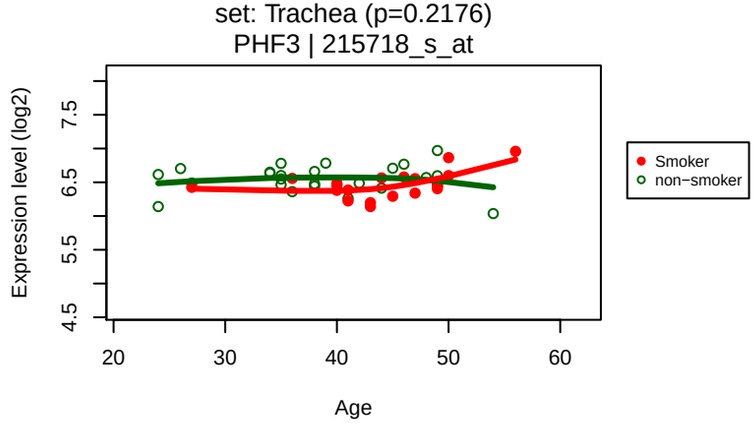

Gene expression ratio in Males
gene: PHF3 | 215718_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5508
p=0.3085
p=0.0544
p=0.1678
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHF3 | 215718_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.082
p=0.0804
p=0.8043
p=0.5653
SmallAirway2
LargeAirway2
Trachea
Macrophage2

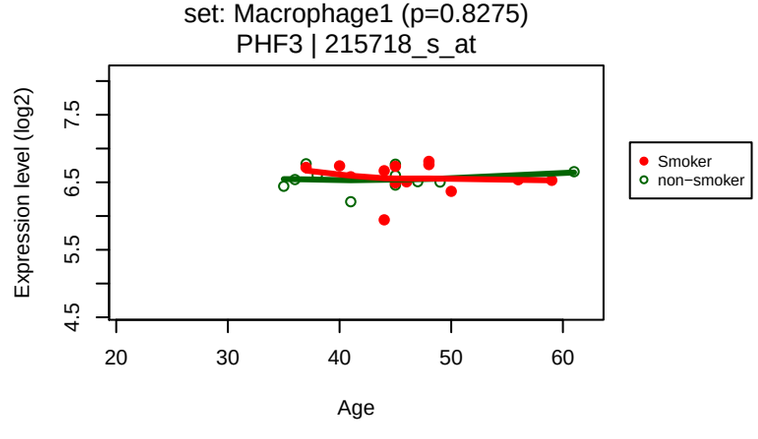

set: SmallAirway1 (p=0.6703)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

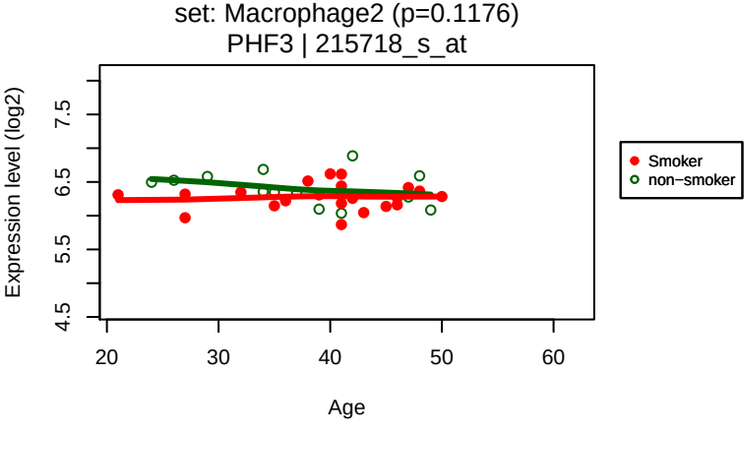

set: SmallAirway2 (p=0.2612)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

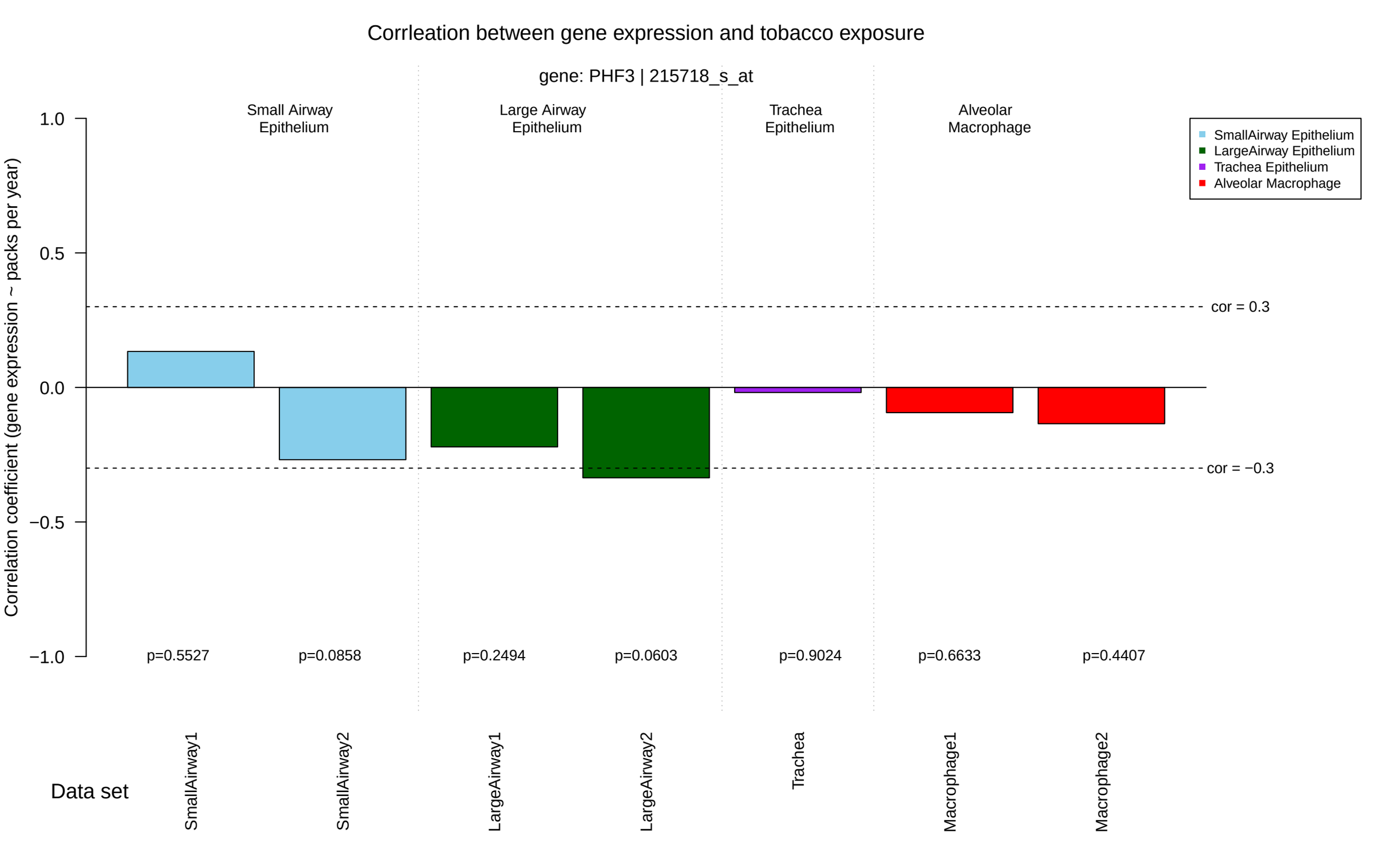

set: LargeAirway1 (p=0.0281)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0336)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2176)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8275)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1176)
PHF3 | 215718_s_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHF3 | 215718_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5527
p=0.0858
p=0.2494
p=0.0603
p=0.9024
p=0.6633
p=0.4407
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2